Pendulums, Driving, and Chaos: Exercise Guide for 2B

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guides

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This is a collection of notes from my Mathematical Methods for Physicists II coursework on free, damped, and driven pendulums. It accompanies notebook 2B. It explains the concepts behind each experiment and the reasoning needed to implement it independently. The shared foundations guide should be read first.

1 How to use this guide

This guide follows all exercise sections of 2B notebook. Read the shared foundation for the pendulum equation, reduced energy, Euler–Cromer, sensitive dependence, refinement, and Poincaré maps. The two plots in section 7 are explained separately here: first step-size checks, then stroboscopic sampling.

The original figure labels come from Giordano and Nakanishi, Chapter 3. They identify the coursework scenarios; they are not a promise that every displayed curve exactly reproduces a book figure.

2 Build the common pendulum solver

2.1 State, parameters, and initial data

The equations are

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -(g/L)\sin\theta - q\omega + F_D \sin(\Omega_D t).$$

The function arguments specify $g, L, q, F_D, \Omega_D, \theta_0, h, T$ and whether to wrap the angle. All current experiments start with $\omega_0 = 0$. To study a different initial velocity, add an explicit argument rather than silently changing the common solver.

Store the initial time, angle, and angular velocity. Set $N = \text{round}(T/h)$; the exercise choices give the intended endpoints. In a general application verify that this choice really gives the requested duration.

2.2 Understand the update order

At step i, set $t = ih$ and calculate the acceleration from the old state. Update ω first, then update θ using that new value. This is Euler–Cromer, not forward Euler:

$$\omega_{i+1} = \omega_i + h a_i, \qquad \theta_{i+1} = \theta_i + h\omega_{i+1}.$$

If wrapping is requested, map the angle into $[-\pi, \pi)$ before storing it. Append time $(i+1)h$ and both state components. For physical trajectory separation and Poincaré sampling, keep integration unwrapped and wrap only the values used for display.

2.3 Know what the model's coefficients mean

The damping term is proportional to angular velocity. The drive amplitude is angular acceleration, not a force in newtons. With radians dimensionless, q and Ω_D have units s^{-1} and F_D units s^{-2}. The sine's time argument must be in radians. The restoring term uses g/L; setting $g = L = 9.8$ makes this ratio one in the chosen SI setup.

3 Exercise 1: an undamped pendulum

Use $g = 9.8$, $L = 1$, $q = F_D = 0$, $\theta_0 = 0.2$, $h = 0.02$, and $T = 10$. At this small angle, the approximate frequency is $\sqrt{9.8}$ and period $2\pi/\sqrt{9.8} \approx 2.01$ s.

Run the common solver, plot angle and angular velocity against time, then plot ω against θ. The initial velocity is zero, and the initial acceleration is $-9.8 \sin 0.2$. Angle and velocity have different units even when displayed together. Their oscillations are approximately a quarter cycle out of phase.

The undamped phase curve should be approximately closed. Energy fluctuations from Euler–Cromer are numerical. The small-angle sinusoid is a useful approximation, but the solver uses the full nonlinear sine, so agreement need not be exact.

4 Exercise 2: damping regimes

Keep the same initial state, g, L, h, T, and compare $q = 1, 5, 10$ with no drive. For each case make an angle-time plot and a phase portrait.

The small-angle critical value is $q_c = 2\sqrt{9.8} \approx 6.26\ \mathrm{s}^{-1}$. Thus the first two cases are underdamped; the last is overdamped. An underdamped response spirals toward equilibrium in phase space. The overdamped response can approach without repeated oscillations.

The exact physical energy satisfies $\dot{\mathcal{E}} = -q\omega^2 \leq 0$. Numerical point-by-point behaviour may have errors, but a sustained large artificial energy increase would be suspicious. Strong damping does not necessarily produce the fastest late return: the slow overdamped rate tends to Ω_0^2/q.

5 Exercise 3: a weak drive and large-amplitude librations

The parameter tuples $(q, \theta_0, F_D, \Omega_D)$ are

$$(1, 0.2, 0.2, 2), \quad (0, 2.5, 0, 2), \quad (0, 1.5, 0, 2),$$

with $g = 9.8, L = 1, h = 0.02, T = 10$. Iterate over these tuples and plot each angle history and phase trajectory.

The first case includes damping and driving; distinguish its initial transient from the later forced response. The other cases are unforced, so their frequency argument has no effect. Both begin at rest below the upright angle π and therefore have energies below the separatrix. They librate rather than continually rotate in the exact model.

At these large initial angles, do not replace $\sin\theta$ by θ. The larger-amplitude free pendulum has a longer period and a visibly non-elliptical phase curve. These are nonlinear effects, not evidence for chaos.

6 Exercise 4: vary the periodic drive

Use $g = L = 9.8$, $q = 0.5$, $\Omega_D = 2/3$, $\theta_0 = 0.2$, $h = 0.04$, $T = 60$, and $F_D = 0, 0.5, 1.2$. The drive period is $T_D = 3\pi \approx 9.42$ s, so this run covers only about 6.37 driving cycles.

For each amplitude, integrate with wrapping enabled and plot angle and velocity separately against time. The $F_D = 0$ response loses energy through damping. With a drive, the input and dissipation can balance in sustained motion, whose form depends on the nonlinear dynamics.

Wrapped-angle jumps at $\pm\pi$ are representation changes. They do not mean the pendulum teleports or acquires infinite angular velocity. An irregular short time series is a motivation for further diagnostics; it does not establish chaos by itself. The extension later examines substantially longer motion at fixed driving phase.

7 Exercise 5: nearby initial conditions

7.1 What is changed and what is held fixed

Compare $\theta_0 = 0.200$ and 0.199, both at rest, with identical parameters and drive phase. Use $g = L = 9.8$, $q = 0.5$, $\Omega_D = 2/3$, and $h = 0.04$. Run $F_D = 0.5$ for 50 s and $F_D = 1.2$ for 150 s.

This changes the physical initial condition by 0.001 rad. It is a different experiment from changing the solver step for the same initial condition, although both can reveal amplification of small differences.

7.2 Build the differences

Integrate both trajectories without wrapping. At each saved time calculate

$$d_{\mathrm{raw}} = |\theta_A - \theta_B|, \qquad d_\theta = |((\theta_A - \theta_B + \pi) \bmod 2\pi) - \pi|.$$

The raw difference retains relative rotation count. The circular difference describes current orientations and is bounded by π. Overlay both on the left panel, then wrap each angle separately for the phase portraits on the right.

Two orientations can coincide while velocities differ. A large unwrapped difference can represent extra revolutions rather than a large current orientation gap. Use these distinctions before interpreting the curves as sensitivity.

7.3 What the evidence can establish

Growing separation supports sensitivity but does not by itself measure the largest Lyapunov exponent or prove a chaotic attractor. Transients, different attractor basins, and numerical errors can complicate the comparison. Periodic driven responses can eventually approach the same motion when both initial states lie in its basin. A chaotic response can remain on the same attractor while individual trajectories cease matching.

8 Exercise 6: integration methods and reduced energy

This later addition compares Euler–Cromer, midpoint RK2, and classical RK4 for the same undamped, undriven pendulum with $g/L = 1$, $\theta_0 = 0.2$, $\omega_0 = 0$, $h = 0.1$, and 800 steps ($T = 80$).

For RK2 and RK4 use the coupled state (θ, ω) and acceleration $-\sin\theta$. RK stages use intermediate angles and velocities. The Euler–Cromer result comes from the original solver; the other two use the separate comparison function.

For every state calculate

$$\mathcal{E} = \omega^2/2 + 1 - \cos\theta, \qquad \Delta\mathcal{E} = \mathcal{E}(t) - \mathcal{E}(0).$$

Plot angle and reduced-energy change, and print $\max|\Delta\mathcal{E}|$. The energy label now correctly uses s^{-2}. Multiply by mL^2 to recover joules; the notebook does not specify those factors in this normalized comparison.

Euler–Cromer can show bounded oscillatory energy error, RK2 can accumulate drift, and RK4 is accurate for this step and duration. The conservation law applies because both drive and damping vanish. It would be inappropriate to demand constant energy in the earlier driven or damped exercises.

9 Extension A: repeat sensitivity at smaller steps

9.1 The new scientific question

Exercise 5 asks whether two nearby physical initial states remain close at one chosen resolution. The extension asks whether that interpretation survives changing the numerical resolution. Its purpose is to detect a conclusion that depends strongly on the chosen discrete update rather than the continuous pendulum equation.

A solver with step h constructs a discrete map. Its truncation errors can alter periods, energy transfer, or attractors. Therefore a visually complicated trajectory at one step size is not enough. Refinement is a test of the simulation's reliability, not a new source of physics.

9.2 Reconstruct the nested-loop logic

For drive 0.5 use the original 50 s duration; for drive 1.2 use 150 s. Within each case repeat at $h = 0.04, 0.02, 0.01$.

1. Choose the drive and duration for the panel.
2. Choose a step size and run both initial angles at that step.
3. Convert the two angle lists to arrays and compute circular separation element by element.
4. Plot separation against that run's own time array on a logarithmic vertical scale.
5. Repeat with the smaller steps while keeping all physical inputs fixed.

The two trajectories within a pair share the same time grid. Between resolutions, the grids differ; the code overlays the curves but does not subtract them point by point. A direct cross-resolution subtraction would

require matching times, for example comparing every second fine-grid state with the coarse grid.

9.3 Why exact long-time agreement is the wrong criterion

In a chaotic regime, truncation and roundoff perturbations can grow along with a deliberate initial-angle perturbation. Schematically, an error of size ϵ amplified at rate λ becomes $\epsilon e^{\lambda t}$. Its tolerance-limited prediction time grows logarithmically as the initial error is reduced.

Consequently two accurate resolutions can eventually give different detailed trajectories. That does not mean refinement has failed. It means a pointwise long-term prediction needs increasingly precise initial data and integration. Conversely, immediate disagreement at short times can indicate inadequate resolution.

Look for agreement during the early resolved evolution and for persistence of the broad regular-versus-sensitive behaviour. Examine the Poincaré patterns as a complementary longer-term diagnostic. Neither similar separation curves nor similar pictures proves full numerical convergence.

9.4 What a semilog plot means

If a suitable distance truly behaves as $d_0 e^{\lambda t}$, then $\log d$ is approximately linear in time over its linearized growth interval. Here the plotted distance is only the angular component. It can dip when orientations align, even when angular velocities differ, and saturates at π because it is a shortest angular distance.

The code plots $\max(d_\theta, 10^{-15})$ to avoid taking a logarithm of zero. The floor is a display device, not a measurement. Do not interpret it as physical saturation or extract an exponent by fitting the entire curve. Transient growth, saturation, wrapping, and projection all affect its shape.

9.5 A further check that is not implemented

For a quantitative Lyapunov exponent, evolve the variational equations

$$\delta\dot{\theta} = \delta\omega, \qquad \delta\dot{\omega} = -\cos\theta(t)\,\delta\theta - 0.5\delta\omega$$

for these normalized parameters, with the same drive phase. Periodically renormalize the perturbation and accumulate its logarithmic growth after discarding transients. Repeated renormalization keeps it small enough for linearization. Test the exponent's convergence with step size and averaging duration.

That procedure is additional work; the present notebook intentionally provides qualitative sensitivity and refinement plots instead. Keeping this distinction prevents overinterpreting its output.

10 Extension B: construct a Poincaré section

10.1 Why the continuous phase portrait is hard to read

A phase portrait displays the pendulum at every phase of the forcing cycle. Regular motion can make complicated loops, and chaotic motion can fill overlapping paths. States with the same (θ, ω) but different drive phases have different future accelerations, so these overlaid paths omit part of the state information.

Introduce $\phi = \Omega_D t \bmod 2\pi$. The extended state is (θ, ω, ϕ). Looking only when $\phi = 0$ removes the drive-phase ambiguity and defines a two-dimensional stroboscopic map. This is the conceptual reason to sample at an exact driving period.

10.2 Derive the sampling schedule

For $\Omega_D = 2/3$,

$$T_D = 2\pi/(2/3) = 3\pi.$$

The extension chooses $M = 240, 480, 960$ steps per period and defines $h = T_D/M$. The corresponding steps are approximately 0.03927, 0.01963, and 0.00982 s. These are close to, but distinct from, the 0.04, 0.02, and 0.01 values in Extension A.

Choosing h this way makes the desired sample indices integers. With index zero holding the initial state, period n ends at index nM. The code integrates for 400 periods, approximately 3769.91 s, discards the first 100 periods, and retains indices

$$101M, 102M, \ldots, 400M.$$

This gives exactly 300 samples at the same drive phase. The first retained sample is at about 951.90 s; the period-100 endpoint is not included.

If one instead rounded $T_D/0.04$ to a stride and sampled repeatedly at that stride, each snapshot would occur slightly early or late. The phase discrepancy would accumulate. Exact period-aligned stepping avoids this source of a falsely smeared section.

10.3 Translate the schedule into code logic

For each drive strength 0.5 and 1.2, and for each M:

1. Calculate $h = T_D/M$ and total duration $400T_D$.
2. Start the pendulum at angle 0.2 and zero velocity, with $q = 0.5$ and $g/L = 1$.
3. Integrate unwrapped histories of angle and velocity.
4. Build the integer index array from 101 to 400, multiplied by M.
5. Select both angle and velocity at these identical indices.
6. Wrap only the selected angles into $[-\pi, \pi)$.
7. Scatter-plot angle against velocity without connecting the points.

The selected velocity must come from the same time as the selected angle. Connecting points with lines would suggest continuous paths that are not the intended section. Keeping matched plot limits across resolutions helps compare shape; ensure limits do not hide relevant samples when changing parameters.

10.4 Read a point, a cycle, or a distributed set

For a stable response repeating every T_D, all late photographs converge to one point. A response repeating every rT_D gives r recurring points. The map uses angle modulo 2π, so a rotating pendulum can repeat its sampled orientation and velocity while completing an integer number of turns between photographs.

A smooth closed sampled curve can indicate quasiperiodicity. A structured distributed set can support a chaotic-attractor interpretation. However, a transient or high-period orbit can also produce many points, and insufficient numerical resolution can distort the set. The finite number of samples cannot establish a fractal dimension or guarantee chaos.

10.5 Interpret the revised notebook's output

For drive 0.5, the samples form a compact point-like pattern at all three tested resolutions, consistent with a drive-periodic attractor for this initial condition. For drive 1.2, they form a distributed structured pattern whose overall appearance persists under refinement, supporting the chaotic interpretation together with the sensitivity experiment.

Individual points in the 1.2 sections need not coincide across step sizes; the corresponding trajectories can have separated long before sampling. Compare the persistent organisation of the set rather than requiring the same sequence of samples. The point-like pattern can also shift slightly with step size because the discrete solver changes the periodic orbit.

10.6 Transients and basins deserve their own checks

The first 100 periods are discarded to reduce initial-condition transients. This number is an exercise choice. To strengthen the result, discard more periods, retain a longer sample, and compare early and late retained blocks. A slowly decaying transient can masquerade as a curve or cloud.

A basin is the set of initial states that approach a given attractor. Different starting angles or velocities can approach different coexisting attractors at the same physical parameters. The displayed section describes the trajectory beginning at 0.2 rad and zero velocity; it is not a map of every possible basin.

10.7 Why damping can coexist with chaotic structure

For the exact flow the phase-plane divergence is

$$\frac{\partial \dot{\theta}}{\partial \theta} + \frac{\partial \dot{\omega}}{\partial \omega} = -q.$$

An infinitesimal area therefore contracts over a period by e^{-qT_D}. With $q = 0.5$, this factor is $e^{-1.5\pi} \approx 0.0090$. This does not require every distance to shrink: one direction may stretch while another contracts strongly. The combination of contraction, nonlinear stretching, and repeated return can create a structured dissipative attractor.

This area statement concerns the exact continuous dynamics. A finite-step Euler–Cromer map approximates it and need not reproduce the factor exactly, which is another reason for refinement.

11 Putting the diagnostics together

The original time plots describe the motion; nearby-angle comparisons explore sensitivity; smaller-step repetitions check its dependence on numerical resolution; Poincaré sections reveal recurrence at fixed forcing phase. Each answers a different question.

For an independent implementation, begin with the undriven small-angle limit and check signs and update order. Add damping, then driving. Retain the drive phase consistently, separate wrapped from unwrapped data, and use the reduced-energy definition with its proper units. Only after these checks interpret the longer driven simulations. Quantitative Lyapunov exponents, basin surveys, and attractor statistics would be further investigations beyond the current notebook.

12 Book references and reading route

1. Nicholas J. Giordano and Hisao Nakanishi, *Computational Physics*, 2nd edition, Addison-Wesley (2005), Chapter 3. This is the coursework reference for oscillatory motion and the driven pendulum. The figure numbers in 2B retain the original homework labels; they are not a claim of exact reproduction.
2. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Read the discussions of phase planes, limit cycles, Poincaré maps, and chaos for the dynamical interpretation of 2B. Topic titles are given because page and chapter locations vary by edition.
3. William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd edition, Cambridge University Press (2007), Chapter 17. Sections 17.1–17.4 cover Runge–Kutta, step-size control, extrapolation, and conservative second-order equations. Official contents: https://numerical.recipes/book/.

These notes are an independent explanation of the coursework, not extracts from these books. Begin with Giordano and Nakanishi for computational examples, use Strogatz for conceptual dynamics, and consult Numerical Recipes for numerical error control.